# Yu Guo

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## About Me

My background is mainly focused on **Computer Graphics**, specially in **Physics-based Rendering**, **Inverse-rendering**, and **Material Capture and generation** using **GAN/Diffusion** model. Recently more focusing on **light/shadow** and material properties decomposition from 3D models (Mesh/**3DGS**) or 2D images, further make it **relightable** and **editable**. I am also interested in the projects related to **Meta Human**. See last page for some examples.

## Education

**University of California, Irvine** — **Irvine, CA, US**
*Ph.D in Computer Science* — **Sept. 2016 – Aug. 2021**
**Advisor**: Shuang Zhao

**University of Chinese Academy of Sciences** — **Beijing & Shenzhen, China**
*M.S. in Computer Science* — **Sept. 2010 – Jul. 2013**
**Advisor**: Pheng Ann Heng (CUHK)

**Central South University** — **Changsha, China**
*B.S. in Mathematics and Applied Mathematics* — **Sept. 2006 – Jul. 2010**

## Working Experiences

**Futurewei Technologies** *(Staff Research Engineer)* — **NJ, US**
**Projects**: — **Sept. 2024 – Current**
Physics-based images/videos generation and editing; Fine-tuning LMM for specific tasks.

**Tencent America** *(Senior Researcher)* — **NY & CA, US**
**Projects:** — **Sept. 2021 – Sept. 2024**
- 3DGS relighting; Portrait relighting; Intrinsic image editing.
- Video generation: *Re-stylization and stabilization of rendered MMD model with Stable diffusion.*
- Product image generation: *We use fine-tuned Diffusion model to generate high quality image, and use image-based relighting technique to make the foreground and background lighting consistant.*
- Texture map delighting: *Remove shadows and highlights in texture maps and make Photogrammetry pipeline more efficient.*
- Unreal Engine 5 plug-in: *Volumetric rendering with multiple scattering and phase function supported.*

**Manager**: Changxi Zheng (NY) and Bo Yang (CA)

**Facebook Reality Lab** *(Internship)* — **Sausalito, CA, US**
**Projects**: Eye caustics rendering and its inverse problem. — **July. 2020 – Sept. 2020**
**Advisor**: Christophe Hery, Olivier Maury

**Adobe Research** *(Internship)* — **San Jose, CA, US**
**Projects**: Material capture and estimation. — **July. 2019 – Sept. 2019**
**Advisor**: Miloš Hašan, Kalyan Sunkavalli

**Megvii (Face++) Research** *(Internship)* — **Redmond, WA, US**
**Projects**: Human face shadow/highlight removal and face relighting. — **July. 2018 – Sept. 2018**
**Advisor**: Jue Wang

**Autodesk** *(Internship)* — **San Francisco, CA, US**
**Projects**: Efficient volumetric rendering of 3D-printing materials. — **July. 2017 – Sept. 2017**
**Advisor**: Miloš Hašan

**Nanyang Technological University** — **Singapore**
*Research Associate at BeingThere Centre (BTC), IMI* — **Oct. 2013 – Mar. 2016**
(BTC is a US$18 million international research project on 3D Telepresence and Virtual Reality between ETH (Markus Gross), UNC (Henry Fuchs) and NTU (Nadia Magnenat Thalmann).)
**Projects**: Stereo rendering; Physical-based video manipulation; Virtual try-on system for eye-glasses.
**Collaborators**: Miriam Reiner, Jean-Charles Bazin, Tobias Martin, Claudia Plüss, P.Y Laffont, Qian Zhang
**Advisor**: Tat-Jen Cham

**Shenzhen Institutes of Advanced Technology** — **Shenzhen, China**
*Research Assistant at HCI lab* — **Sept. 2011 – Jul. 2013**
**Projects**: Mesh processing; Soft body simulation; Virtual surgery; CUDA acceleration.
**Advisor**: Pheng-Ann Heng, Yongming Xie

## Selected Publications

(See full publication list in Google Scholar)

"**SIRR-LMM: Single-image Reflection Removal via Large Multimodal Model**" by **Yu Guo**, Zhiqiang Lao, Xiyun Song, Yubin Zhou, Heather Yu. *(WACV 2026)*

"**Fluid Composer: Fluid Detail Composition and Rendering Using Video Diffusion Models**" by Duowen Chen, Zhiqiang Lao, **Yu Guo**, Heather Yu. *(CGF 2025)*

"**ePBR: Extended PBR Materials in Image Synthesis**" by **Yu Guo**, Zhiqiang Lao, Xiyun Song, Yubin Zhou, Zongfang Lin, Heather Yu. *(CVPR 2025)*

"**Seeing A 3D World in A Grain of Sand**" by Yufan Zhang, Yu Ji, **Yu Guo**, Jinwei Ye. *(CVPR 2025)*

"**BiGS: Bidirectional Gaussian Primitives for Relightable 3D Gaussian Splatting**" by Liu Zhenyuan, **Yu Guo**, Xinyuan Li, Bernd Bickel, Ran Zhang. *(3DV 2025)*

"**Textureless Deformable Object Tracking with Invisible Markers**" by Xinyuan Li, **Yu Guo**, Yubei Tu, Yu Ji, Yanchen Liu, Jinwei Ye, Changxi Zheng. *(TPAMI 2024)*

"**Woven Fabric Capture from a Single Photo**" by Wenhua Jin, Beibei Wang, Milos Hasan, **Yu Guo**, Steve Marschner and Lingqi Yan. *(SIGGRAPH Asia 2022)*

"**Beyond Mie Theory: Systematic Computation of Bulk Scattering Parameters based on Microphysical Wave Optics**" by **Yu Guo**, Adrian Jarabo and Shuang Zhao. *(TOG 2021)*

"**MaterialGAN: Reflectance Capture using a Generative SVBRDF Model**" by **Yu Guo**, Cameron Smith, Miloš Hašan, Kalyan Sunkavalli and Shuang Zhao. *(TOG 2020)*

"**A Bayesian Inference Framework for Procedural Material Parameter Estimation**" by **Yu Guo**, Miloš Hašan, Lingqi Yan and Shuang Zhao. *(CGF 2020)*

"**Position-Free Monte Carlo Simulation for Arbitrary Layered BSDFs**" by **Yu Guo**, Miloš Hašan and Shuang Zhao. *(TOG 2018)*

"**A Virtual Try-on System for Prescription Eyeglasses**" by Qian Zhang, **Yu Guo**, Pierre-Yves Laffont, Tobias Martin, and Markus Gross. *(CG&A 2017)*

"**Physically Based Video Editing**" by Jean-Charles Bazin, Claudia Plüss (Kuster), **Yu Guo**, Tobias Martin, Alec Jacobson, and Markus Gross. *(CGF 2016)*

## Reviews

TOG, CGF, SIGGRAPH, SIGGRAPH Asia, EG, PG

# Previous Projects

## Rendering:

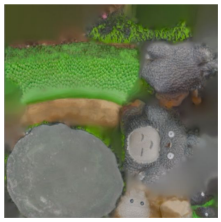
- BSDF
- Layered surface
- PBRT v4

- Wave optics

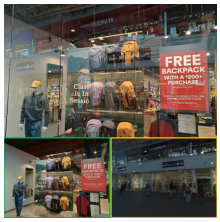
- UE plugin
- Snow rendering
- Multiple scattering

## Inverse Rendering:

- Material recovery

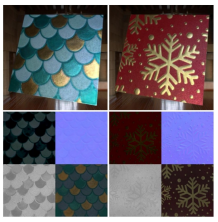


- Photogrammetry
- Shadow removal



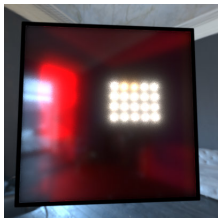

- Reflection removal
- Reflection Separation

## PBR material:



- SVBRDF
- MaterialGAN

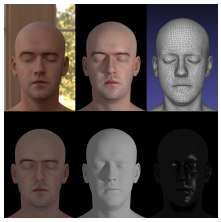
- Procedual material
- MCMC



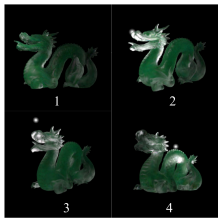
- Extented PBR
- Mirror surface
- Transparent surface

## Delighting and relighting:

- Consumer goods
- Image based



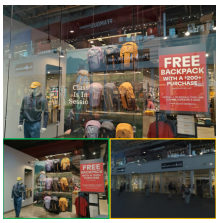
- Human face
- Mesh based




- Translucent object
- 3DGS based

## Generative AI:

- Stable Diffusion
- Fine-tuning
- Dreambooth

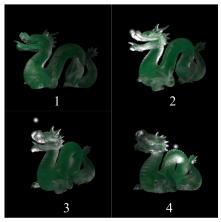



- Flux.1 Kontext
- Fine-tuning
- LoRA

- Video Generation
- Diffusion model

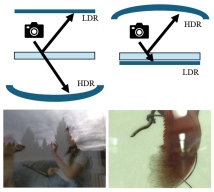
## 3DGS:

- Tiny object

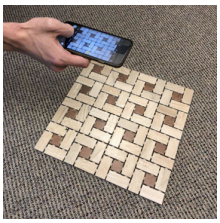



- Translucent object

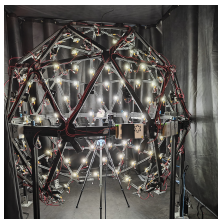
## Data collection:




- Synthetic data



- Portable device



- Light stage